A bound for the number of maximum modulus points

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Abstract

For a non-monomial entire function f, let $\nu_f(r)$ be the number of points on $|z| = r$ at which $|f|$ attains its maximum. Write $f(z) = z^m(c_0 + c_k z^k + O(z^{k+1}))$, where $c_0 c_k \neq 0$ and $k \geq 1$. We prove that $\nu_f(r) \leq 2k$ outside a countable set of radii. In particular, $\nu_f(r)$ cannot tend to infinity, answering the second question of Erdős on maximum modulus points. The proof studies the correspondence $A(z) = \overline{A(\overline{w})}$, where $A = zf'/f$, using implicit-function continuation, growth in a direct tract, and a finite proper map of analytic curves.

1 Introduction

For a nonzero entire function f, put

$$M(r, f) = \max_{|z|=r} |f(z)|, \qquad \nu_f(r) = \#\{z : |z| = r,\ |f(z)| = M(r, f)\}.$$

If f is not a monomial, then $\nu_f(r)$ is finite for every $r > 0$. Erdős asked whether a single such function can satisfy

$$\limsup_{r\to\infty} \nu_f(r) = \infty, \qquad \text{and whether it can satisfy} \qquad \liminf_{r\to\infty} \nu_f(r) = \infty;$$

see [2, 5] for the problem and its history. Herzog and Piranian [7] answered the first question affirmatively. Pardo-Simón and Sixsmith [8] obtained such an example of finite order. Glücksam and Pardo-Simón [5] constructed an approximate analogue of the second property, with many separated arcs on which the modulus is close to its maximum.

There is also a classical local theory near the origin. Hayman [6] proved that for f as in (1.1), the maximum modulus set near zero consists of at most k analytic curves. Evdoridou, Pardo-Simón, and Sixsmith [4] refined this description, determining the exact number of local maximum curves outside an algebraically defined exceptional class and showing that each such curve contains exactly one point of each sufficiently small positive modulus. In particular, $\nu_f(r) \leq k$ for all sufficiently small r.

We prove that the answer to the second question is negative.

Theorem 1.1. *Let f be a nonzero entire function that is not a monomial, and write*

$$f(z) = z^m(c_0 + c_k z^k + O(z^{k+1})), \qquad m \geq 0, \quad k \geq 1, \quad c_0 c_k \neq 0. \tag{1.1}$$

There is a countable set $E \subset (0, \infty)$ such that

$$\nu_f(r) \leq 2k \qquad (r \notin E). \tag{1.2}$$

Consequently,

$$\liminf_{r\to\infty} \nu_f(r) \leq 2k < \infty.$$

The integer k is the difference between the degrees of the first two nonzero terms of f. The factor 2 is attained by an explicit example given in Remark 5.1.

The countable-exception bound allows the unbounded upper limits constructed in [7, 8]: their large counts may occur on the exceptional radii. The separated arcs in [5] concern values close to the maximum, so they need not produce additional points attaining it. The local bound k near the origin from [6, 4] is also consistent with the bound $2k$ for nonexceptional radii here.

Set $A = zf'/f$. At radii where $\log M(r, f)$ is differentiable with respect to $\log r$, all maximum modulus points have the same real value of A. Hence they determine points $(z, \overline{z})$ on the correspondence $A(z) = \overline{A(\overline{w})}$ with the same values of zw and $A(z)$. We show that every irreducible component of this correspondence contains points with $|zw|$ arbitrarily small. Since $A - m$ has local degree k at the origin, the generic fibres of the resulting finite proper map have size at most $2k$.

2 Analytic preliminaries

Lemma 2.1 (Stoïlow; see [3])**.** *Let F be entire on $\mathbb{C}^2$, and suppose that*

$$F(a, b) = 0, \qquad F_w(a, b) \neq 0.$$

The projection to the z-plane of the maximal analytic continuation of the implicit germ $w = w(z)$ through (a, b) is dense in $\mathbb{C}$.

Following Bergweiler, Rippon, and Stallard [1, Definition 2.1], an unbounded domain $D \subset \mathbb{C}$, with unbounded complement and piecewise smooth boundary, is a *direct tract* of Q if Q is holomorphic in D, continuous on its closure, and

$$|Q| = a \quad \text{on } \partial D, \qquad |Q| > a \quad \text{in } D$$

for some $a > 0$.

Lemma 2.2 (Fuchs; see [1, Theorem 2.1])**.** *Let D be a direct tract of Q, with boundary value a, and put*

$$u(w) = \begin{cases} \log(|Q(w)|/a), & w \in D, \\ 0, & w \notin D. \end{cases}$$

Then u is a continuous subharmonic function on $\mathbb{C}$, and

$$\frac{\max_{|w|=R} u(w)}{\log R} \longrightarrow \infty \qquad (R \to \infty). \tag{2.1}$$

3 Small products on the correspondence

Fix f, m, k as in Theorem 1.1. Write $f = z^m h$, where $h(0) \neq 0$, and define the meromorphic functions

$$A(z) = \frac{zf'(z)}{f(z)} = m + \frac{zh'(z)}{h(z)}, \qquad B(w) = A^{\#}(w) := \overline{A(\overline{w})}. \tag{3.1}$$

The removable singularity of A at zero is filled in. The function A is nonconstant, since otherwise f would be a monomial. Moreover, (1.1) gives

$$A(z) - m = \frac{kc_k}{c_0} z^k + O(z^{k+1}).$$

In particular, $k = \mathrm{ord}_0(A - m)$.

There is a holomorphic local coordinate χ at zero such that

$$\chi(0) = 0, \qquad \chi'(0) \neq 0, \qquad A(z) - m = \chi(z)^k. \tag{3.2}$$

Choose $\delta \in (0, 1)$ small enough that

$$V = \chi^{-1}\{\xi : |\xi| < \delta^{1/k}\}$$

has compact closure in the coordinate neighbourhood. For every $|\lambda - m| < \delta$, the equation $A(z) = \lambda$ has exactly k roots in V, counted with multiplicity. They are distinct when $\lambda \neq m$, and there is a constant C such that all of them satisfy

$$|z| \leq C|\lambda - m|^{1/k}. \tag{3.3}$$

The corresponding statements hold for B on $V^{\#} = \{\overline{z} : z \in V\}$.

Choose δ also so that it is not the modulus of any finite, nonzero critical value of $B - m$. Then $\{|B - m| = \delta\}$ is a locally finite union of smooth curves.

Choose entire functions p, q with no common zero such that $A = p/q$ and $q(0) \neq 0$. Set

$$F(z, w) = p(z)q^{\#}(w) - p^{\#}(w)q(z), \qquad \mathcal{X} = \{(z, w) \in \mathbb{C}^2 : F(z, w) = 0\}. \tag{3.4}$$

All analytic sets in this paper are given their reduced structure. Away from poles, the equation in (3.4) is $A(z) = B(w)$. Since p, q have no common zero and A is nonconstant, $\mathcal{X}$ has no horizontal or vertical irreducible component. At points with finite common value,

$$F_w(z, w) = -q(z)q^{\#}(w)B'(w).$$

It follows that $F_w \neq 0$ at a general point of each irreducible component: the coordinate w is nonconstant there, whereas the zeros of B' are discrete.

Lemma 3.1. *Every irreducible component X of $\mathcal{X}$ contains a sequence (z_j, w_j) such that*

$$z_j w_j \neq 0, \qquad A(z_j) = B(w_j) \in \mathbb{C} \setminus \{m\}, \qquad |z_j w_j| \longrightarrow 0. \tag{3.5}$$

Proof. Choose a regular point of X at which $F_w \neq 0$ and apply Lemma 2.1 to the corresponding implicit germ. Since its projection is dense, the continuation reaches a point $(z_0, w_0) \in X$ with

$$z_0 \in V \setminus \{0\}, \qquad 0 < |A(z_0) - m| < \delta.$$

Let U be the connected component of $\{|B - m| < \delta\}$ containing w_0. In $V \times U$, consider the analytic curve

$$\chi(z)^k = B(w) - m. \tag{3.6}$$

Its projection to U is finite and proper. Indeed, over a compact subset of U, the right-hand side remains in a compact subset of $\{\xi : |\xi| < \delta\}$, so all its k-th roots remain in a compact subset of the coordinate disc.

Since $A'(z_0) \neq 0$, the curve (3.6) is smooth at (z_0, w_0). Let Y be its irreducible component through this point. Near (z_0, w_0), it is the same branch as X, hence $Y \subset X$ by the identity theorem for analytic sets. The projection $Y \to U$ is finite, proper, and nonconstant. Its image is therefore both open and closed in U, so the projection is surjective. Thus every $w \in U$ has a lift $(z, w) \in Y$, with z satisfying (3.3).

If $B(b) = m$ for some $b \in U$, choose $w_j \to b$ in U, with $w_j \neq 0$ and $B(w_j) \neq m$, and lift these points to Y. Equation (3.3) gives $z_j \to 0$, and hence $z_j w_j \to 0$. This proves the lemma in this case.

Suppose now that $B - m$ has no zero in U, and put

$$Q(w) = \frac{\delta}{B(w) - m}. \tag{3.7}$$

Every finite boundary point of U satisfies $|B - m| = \delta$. Thus Q is holomorphic in U, continuous on $\overline{U}$, and satisfies

$$|Q| > 1 \quad \text{in } U, \qquad |Q| = 1 \quad \text{on } \partial U.$$

The maximum principle shows that U is unbounded.

We claim that its complement is also unbounded. If f has a zero other than the origin, then for every sufficiently large R whose circle avoids the zeros of f, the argument principle gives

$$\frac{1}{2\pi}\int_0^{2\pi} (B(Re^{i\theta}) - m)\, d\theta = n(R, f) - m \geq 1, \tag{3.8}$$

where $n(R, f)$ counts zeros in $|z| < R$, with multiplicity. Here $B(w) = w(f^{\#})'(w)/f^{\#}(w)$, and f and $f^{\#}$ have the same zero counts on centred discs. Since $\delta < 1$, such a circle cannot be contained in $\{|B - m| < \delta\}$. A circle containing a pole of B is not contained in that set either. If f has no zero other than possibly the origin, then $B - m$ is a nonconstant entire function. Its maximum modulus eventually exceeds δ, giving the same conclusion for all large circles. In either case, $\mathbb{C} \setminus U$ is unbounded.

Consequently, U is a direct tract of Q with boundary value one. Define $u = \log|Q|$ in U, and $u = 0$ outside U. For each sufficiently large R, choose w_R on $|w| = R$ at which u attains its maximum. By Lemma 2.2, this maximum is positive and

$$\frac{u(w_R)}{\log R} \longrightarrow \infty.$$

In particular, $w_R \in U$. Choose a lift $(z_R, w_R) \in Y$. Equations (3.3) and (3.7) yield

$$|z_R w_R| \leq C\delta^{1/k} R \exp\Big(-\frac{u(w_R)}{k}\Big) \longrightarrow 0. \tag{3.9}$$

Since $B - m$ has no zero in U, these lifts have nonzero coordinates and common value different from m. They satisfy (3.5). □

4 A finite map of analytic curves

Remove the zero coordinates, poles, and common value m, and set

$$\begin{aligned}\mathcal{X}^* &= \{(z, w) \in \mathcal{X} : zw \neq 0,\ A(z) = B(w) \in \mathbb{C} \setminus \{m\}\},\\ \mathcal{B} &= \mathbb{C}^* \times (\mathbb{C} \setminus \{m\}).\end{aligned}$$

On each irreducible component of $\mathcal{X}$, the deleted locus is discrete. Thus every irreducible component of $\mathcal{X}^*$ is obtained from one of $\mathcal{X}$ by deleting a discrete set. Consider the holomorphic map

$$\Phi : \mathcal{X}^* \longrightarrow \mathcal{B}, \qquad \Phi(z, w) = (zw, A(z)) = (s, \lambda). \tag{4.1}$$

Lemma 4.1. *The map Φ is proper and has finite fibres.*

Proof. Suppose that (s, λ) remains in a compact subset of $\mathcal{B}$. Then $|s|$ is bounded above and away from zero, while λ is bounded and stays a positive distance from m. If a sequence of preimages had $z \to \infty$, then $w = s/z \to 0$, and therefore $\lambda = B(w) \to m$, a contradiction. If $z \to 0$, then $\lambda = A(z) \to m$, again a contradiction. The same arguments apply with z and w interchanged. Neither coordinate can approach a pole, since λ is bounded. It follows that the inverse image of the compact set is compact in $\mathcal{X}^*$.

For fixed (s, λ), the possible z-coordinates form a compact subset of the discrete zero set of $A - \lambda$. There are therefore only finitely many of them, and each determines $w = s/z$ uniquely. $\square$

Proposition 4.2. *The image $\Gamma = \Phi(\mathcal{X}^*)$ is a closed analytic curve in $\mathcal{B}$. There is a locally finite set $T \subset \Gamma$ such that*

$$\#\Phi^{-1}(s, \lambda) \leq 2k \qquad ((s, \lambda) \in \Gamma \setminus T). \tag{4.2}$$

Proof. By Remmert's proper mapping theorem and Lemma 4.1, Γ is a closed analytic curve. Let T consist of the singular points of Γ together with the images of the singular and ramification points of $\mathcal{X}^*$, where ramification is understood on the normalization of each source component. The map is nonconstant on every source component, so its ramification locus is discrete; properness then makes these exceptional sets locally finite in Γ.

For each irreducible component Γ_i of Γ, the map

$$\Phi^{-1}(\Gamma_i \setminus T) \longrightarrow \Gamma_i \setminus T$$

is a finite covering. The normalization of Γ_i is a connected Riemann surface, and deleting a locally finite set leaves it connected. Since T contains the singular points of Γ, it follows that $\Gamma_i \setminus T$ is connected. Hence there is an integer d_i such that

$$\#\Phi^{-1}(s, \lambda) = d_i \qquad ((s, \lambda) \in \Gamma_i \setminus T). \tag{4.3}$$

Choose $\eta > 0$ so that the disc $\{|z| < \eta\}$ is contained in both V and $V^{\#}$. If $0 < |s| < \eta^2$, every preimage satisfies

$$\min\{|z|, |w|\} < \eta.$$

There are at most k preimages with $|z| < \eta$: the existence of such a coordinate implies $|\lambda - m| < \delta$, and the local root count in Section 3 applies. Each root determines the other coordinate from $w = s/z$. Likewise there are at most k preimages with $|w| < \eta$. Hence

$$\#\Phi^{-1}(s, \lambda) \leq 2k \qquad (0 < |s| < \eta^2). \tag{4.4}$$

The image of each irreducible component of $\mathcal{X}^*$ is an irreducible component of Γ, and every component of Γ arises in this way. Lemma 3.1 therefore shows that each Γ_i meets $\{0 < |s| < \eta^2\}$. This intersection is a nonempty open subset of a curve, so it is not contained in T. Evaluating (4.3) at a point of this intersection and using (4.4) gives $d_i \leq 2k$. Equation (4.2) follows for every component. $\square$

5 Maximum modulus points

Proof of Theorem 1.1. Set

$$v(x) = \log M(e^x, f), \qquad x \in \mathbb{R}.$$

Hadamard's three-circles theorem makes v convex. Its set N of nondifferentiability points is therefore at most countable. Suppose that $x \notin N$, let $r = e^x$, and let $z = re^{i\theta}$ be a maximum modulus point. Since $f(z) \neq 0$, angular differentiation gives

$$0 = \frac{\partial}{\partial \theta} \log |f(re^{i\theta})| = -\operatorname{Im} A(z). \tag{5.1}$$

For this fixed θ, the function $t \mapsto \log |f(e^{t+i\theta})|$ lies below $v(t)$ and agrees with it at $t = x$. Differentiating at this contact point yields

$$\operatorname{Re} A(z) = v'(x). \tag{5.2}$$

Thus all maximum modulus points on the circle have the same real value $A(z) = \lambda := v'(x)$.

In fact $\lambda > m$. To see this, write

$$b(x) = v(x) - mx = \log M(e^x, h).$$

The function b is convex and strictly increasing, since the maximum modulus of a nonconstant entire function strictly increases with radius. For any $t < x$, the convex secant-slope inequality gives

$$0 < \frac{b(x) - b(t)}{x - t} \leq b'(x).$$

Hence $\lambda = m + b'(x) > m$.

For every maximum modulus point z on $|z| = r$, we now have

$$(z, \overline{z}) \in \mathcal{X}^*, \qquad \Phi(z, \overline{z}) = (r^2, \lambda). \tag{5.3}$$

Distinct maximum points give distinct points of this fibre. If $(r^2, \lambda) \notin T$, Proposition 4.2 gives $\nu_f(r) \leq 2k$.

Since T is locally finite, it is countable. Define

$$E = \{e^x : x \in N\} \ \cup \ \{\sqrt{s} : s > 0 \text{ and } (s, \lambda) \in T \text{ for some } \lambda\}.$$

This is a countable set of positive radii, and (1.2) holds for every $r \notin E$. Since every interval (R, ∞) contains radii outside E, the asserted bound on the lower limit follows. □

Remark 5.1. For $k \geq 1$, the function $f(z) = \exp(2z^k - z^{2k})$ has first two nonzero Taylor terms $1 + 2z^k$. Put $R = r^k$ and $\phi = k\theta$. Then

$$\log |f(re^{i\theta})| = R^2 + \frac{1}{2} - 2R^2 \left(\cos \phi - \frac{1}{2R}\right)^2.$$

For $0 < R \leq 1/2$, the maximum is attained when $\cos(k\theta) = 1$, giving k distinct points. For $R > 1/2$, it is attained precisely when $\cos(k\theta) = 1/(2R)$, giving $2k$ distinct points. Thus the bound in Theorem 1.1 is sharp for every k.

Use of generative AI

GPT-6 Astra was used to generate the mathematical proofs and draft the manuscript. GPT-5.6 Sol and Claude Opus 5 were used only for editorial review of the exposition and did not contribute to the mathematical arguments. The author reviewed the final manuscript and takes full responsibility for its content.

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